

Ex Show that if $A = A^T$ then the eigenvalues of A are real

Ex Show that if $A = A^T$ and $\bar{x}_1, \bar{x}_2$ are eigenvectors of eigenvalue $\lambda_1 \neq \lambda_2$ then $\bar{x}_1 \cdot \bar{x}_2 = 0$

Ex Find a basis of orthonormal eigenvectors

for $\begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & 0 \end{pmatrix}$ & for $\begin{pmatrix} 2 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}$

Ex Find a matrix that has a basis of eigenvectors which are not ON.

Ex Compute $\exp \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$ & solve $\begin{cases} \frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ x(0) = 1 \\ y(0) = 2 \end{cases}$

Ex Consider the sequence 0 1 2 5 12 $t_{k+2} = 2t_{k+1} + t_k$

Find A such that $\begin{pmatrix} t_{k+2} \\ t_{k+1} \end{pmatrix} = A \begin{pmatrix} t_{k+1} \\ t_k \end{pmatrix}$

Compute the eigenvalues & eigenvectors of A

Give a formula for t_k .

Ex Explain why if $A \in M_{nn}(\mathbb{R})$ has n -distinct eigenvalues, then it can be diagonalized.

Prove it for $n=2$.